HRT · ETH ZÜRICH DATATHON 2026

Predicting Second-Half Returns

A simple, leak-aware strategy built on three price insights and one honest headline signal.

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THE TASK

Given the first half of a trading session, pick a position for the second half.

1,000 training sessions · 10,000 test · scored by Sharpe × 16. Inputs: 1-minute bars + templated headlines.

POSITION RANGE

[0.2, 2.0]

long-only, clipped

OBSERVED

H1

bars + headlines

PREDICT

H2 P&L

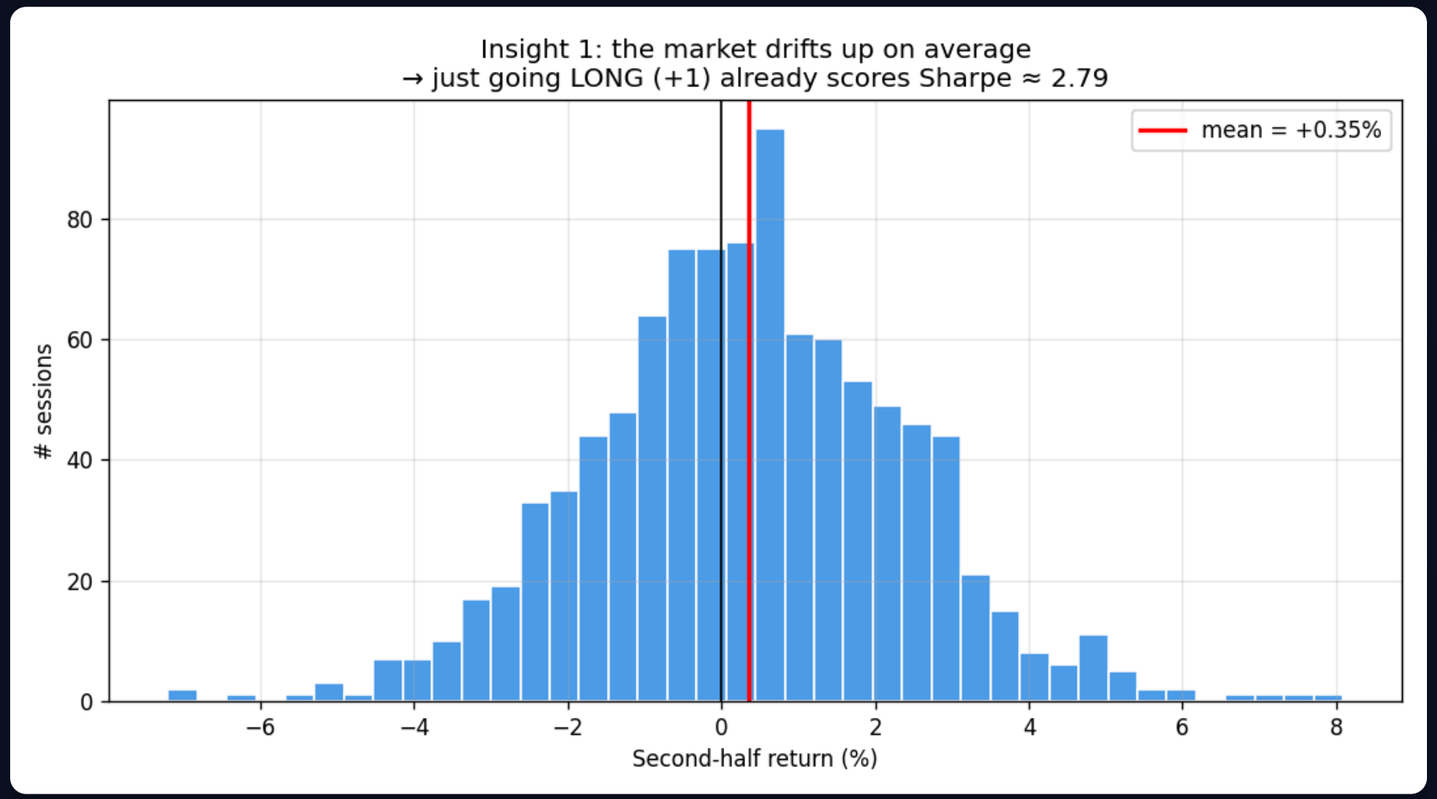
position × sh_ret

```
target_position = f(fh_ret, headlines)
```

INSIGHT 1 · DRIFT

The market drifts up. Going LONG already wins.

H2 mean +0.35%, std 2.0%. Constant position = +1 gives CV Sharpe ≈ 2.79.



H2 MEAN RETURN

+0.35%

H2 STD

2.0%

CONSTANT +1 SHARPE

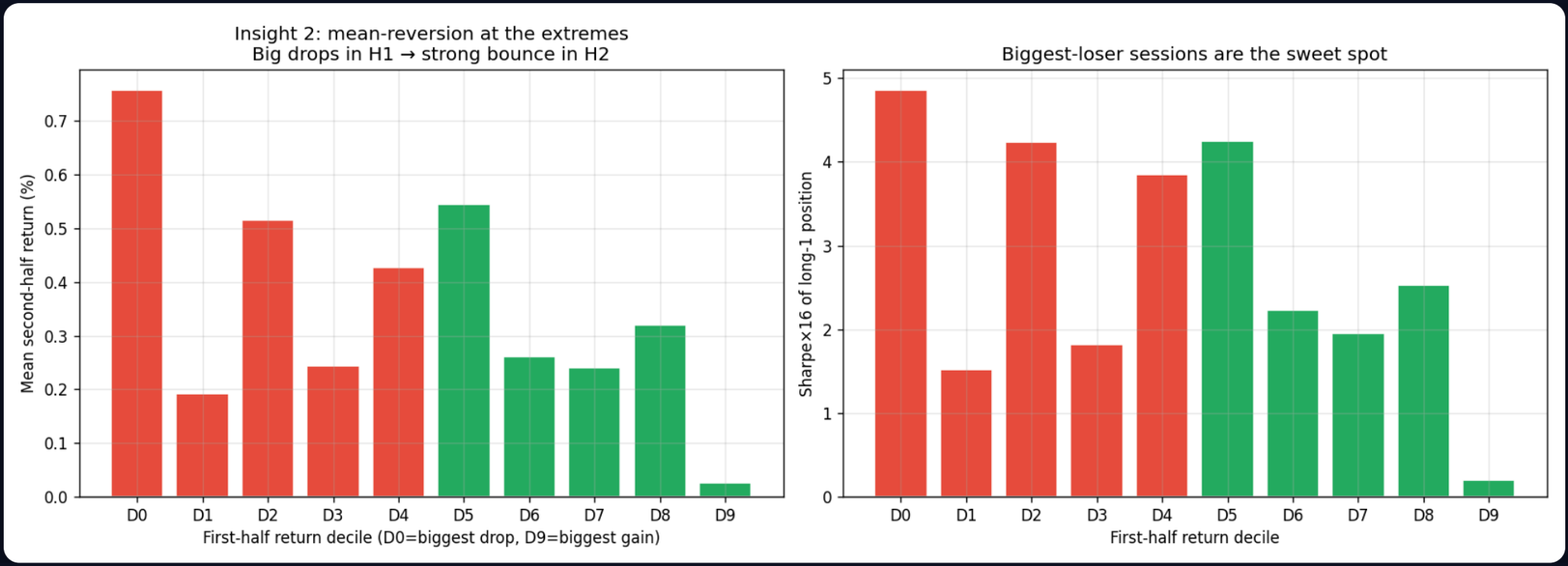
2.79

the baseline to beat

INSIGHT 2 · MEAN REVERSION

Big losers bounce back. Big winners don't keep winning.

If this were momentum, D9 would be the sweet spot — but D0 is.



DECILE	FH_RET	H2 MEAN
D0	−7.0% to −2.4% (big drops)	+0.76%
D4	roughly flat	+0.43%
D9	+2.8% to +7.5% (big gains)	+0.03%

POSITION RULE

$$pos = 1 - 10 \cdot fh_ret$$

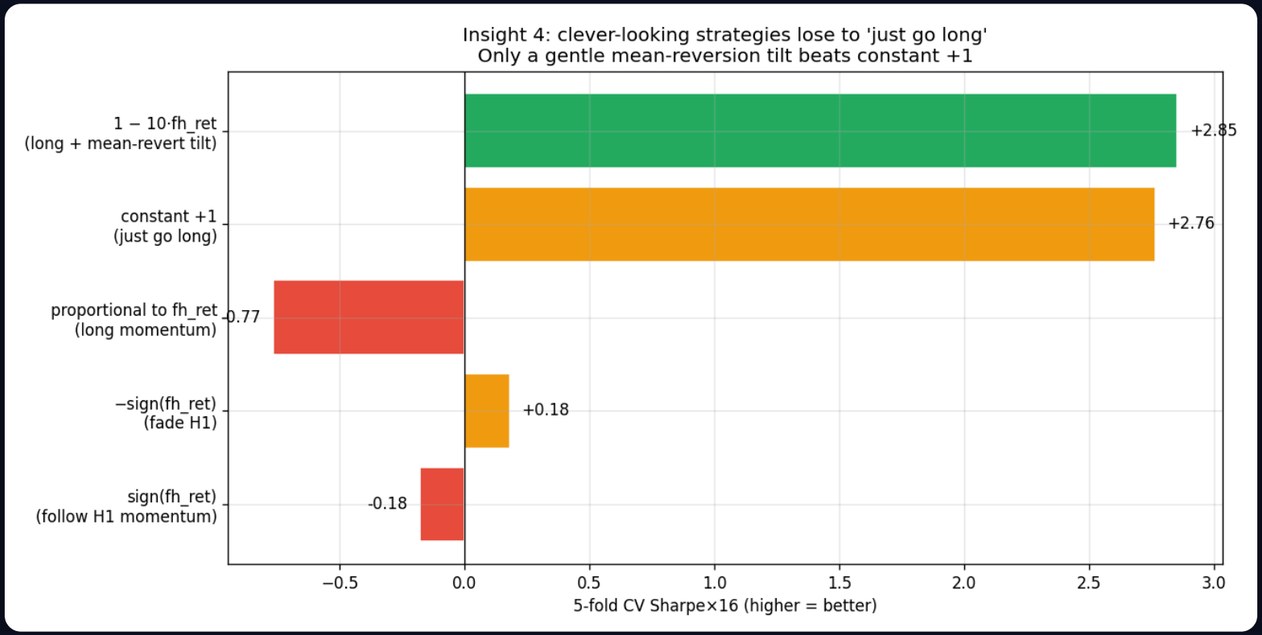
Gentle mean-revert tilt · CV Sharpe ≈ 2.85 (+0.06)

INSIGHT 3 · WHAT BEATS THE BASELINE

Clever direction-picking loses. Only a gentle mean-revert tilt wins.

5-fold CV Sharpe × 16.

STRATEGY	SHARPE
follow H1 momentum (sign(fh_ret))	−0.18
proportional to fh_ret	−0.77
constant +1 (LONG)	+2.76
1 – 10·fh_ret (mean-revert tilt)	+2.85



HEADLINES

Headlines are Mad Libs.

~74 templates, 50 cover 94% of the corpus. Bag-of-words finds nothing after Bonferroni.

```
1,460 × <CO> secures $<N>M contract with <PARTNER>
252 ×   <CO> begins scheduled maintenance of <SECTOR>
235 ×   <CO> reports record quarterly revenue, up <PCT>%
230 ×   <CO> explores strategic alternatives for <SECTOR>
226 ×   <CO> sees <PCT>% drop in new customer orders
224 ×   <CO> completes strategic acquisition

…
```

TEMPLATES

~74

COMPANIES

80

INDIVIDUAL WORDS

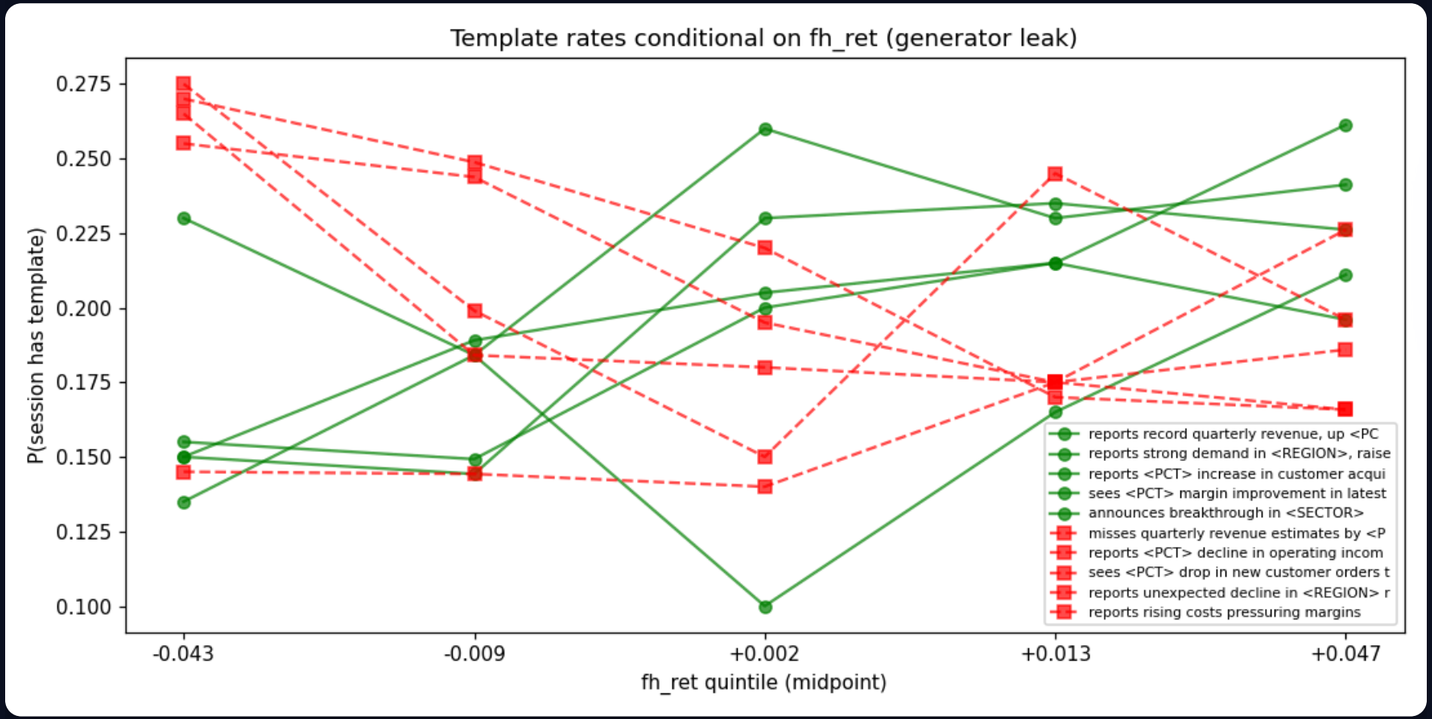
noise

fails Bonferroni

THE TRAP

Templates are conditioned on the price path.

When H1 rallies the generator emits bullish templates. Counting them re-discovers fh_ret.



BMB ↔ FH_RET

r = +0.29

leakage — not independent signal

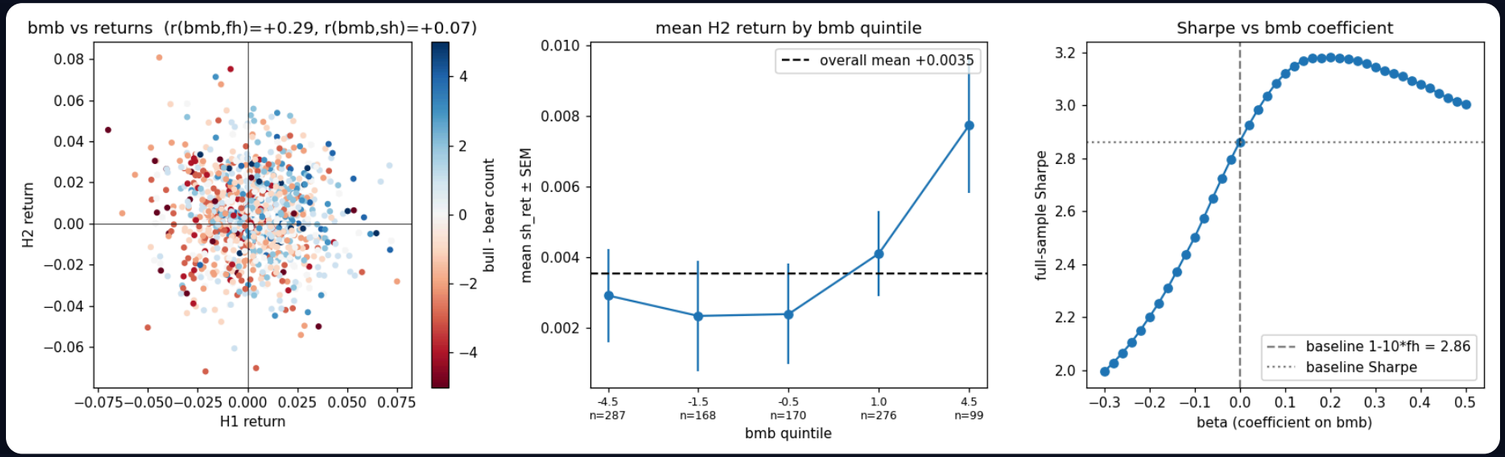
FIX

Residualize fh_ret out of bmb. What remains is the part of headlines that is not already encoded in price.

REAL HEADLINE SIGNAL

After residualizing, bullish templates predict positive H2.

Small but stable, orthogonal to price. News is honest once you control for the path.



RESIDUAL BMB ↔ SH_RET

r = +0.094

p = 0.003

HAND-LABELED TEMPLATES

14 + 14

bull / bear · verified in test vocab

UPLIFT ON BASELINE

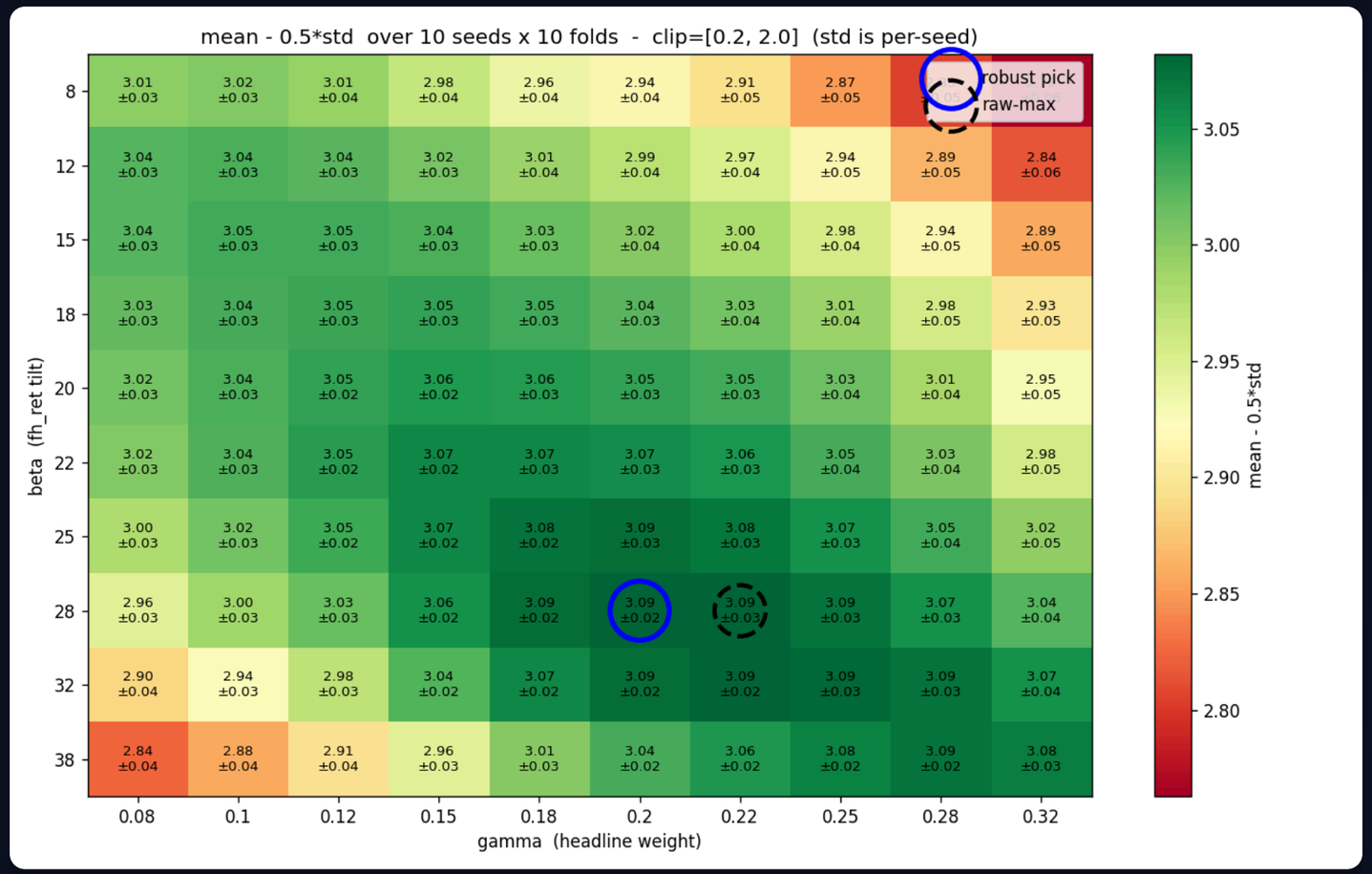
+0.28 Sharpe

improves every fold

FINAL STRATEGY

clip(1 – 28·fh_ret + 0.2·bmb , 0.2 , 2.0)

Long baseline + mean-reversion tilt + residualized headline score. Grid-searched over 10 seeds × 10 folds.



B (FH_RET WEIGHT)

28

robust pick, mean – 0.5·std

Γ (BMB WEIGHT)

0.20

PLATEAU STD / SEED

±0.02–0.04

ranking is reliable

BROAD PLATEAU

β∈[18,32]

γ∈[0.18, 0.25]

RESULTS

CV 3.10 → Public 2.72

A ~0.34 gap. Plateau is broad, so this is small-train variance / mild distribution shift, not cell-luck.

CONSTANT +1

2.76

baseline

+ MEAN-REVERT TILT

2.85

1 – 10·fh_ret

+ HEADLINES (FINAL)

3.12

CV Sharpe × 16

SUBMISSIONS

V	FORMULA	CV	PUBLIC
v3	clip(1 – 28·fh_ret + 0.2·bmb, 0.2, 2.0)	3.10	2.72180

TAKEAWAYS

Three price facts. One leak. One honest signal.

- The market drifts up — start from LONG.
- Fade H1 extremes — mean reversion, not momentum.
- Headlines leak the price path — residualize before trusting them.
- Hand-labeled bull/bear templates give a stable +0.28 Sharpe uplift.

Open: CV→public gap (0.34), and whether embedding-based sentiment beats hand labels.